RoboRide

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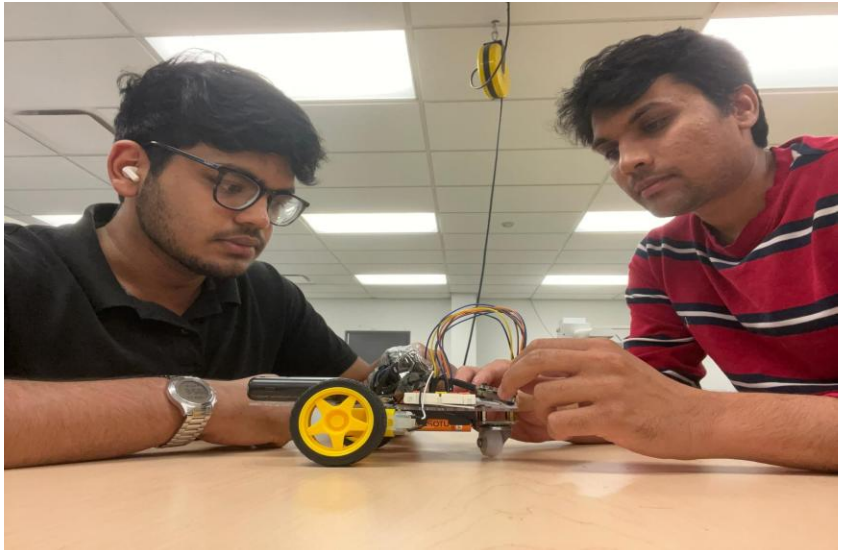
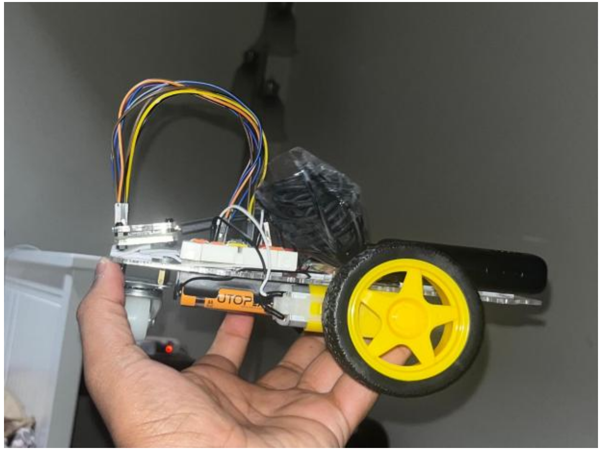
Introduction

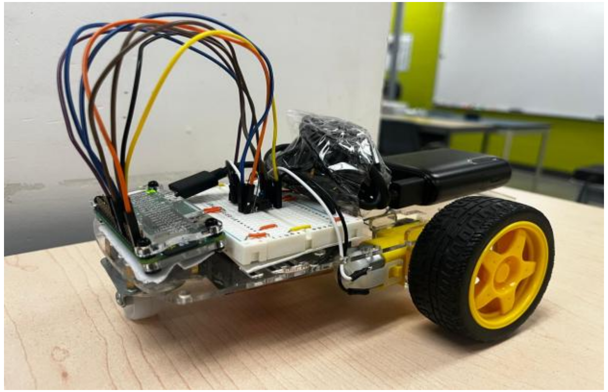
- In a world moving toward smarter automation, the next generation of robotics must not only move it must observe, sense, and make decisions. RoboRide Phase 2 brings our platform one step closer to that future.
- With the addition of an ultrasonic distance sensor and a high-resolution Raspberry Pi camera module, RoboRide has evolved from a simple motorized vehicle into an intelligent, perception-driven robotic system. It can detect obstacles, protect itself while reversing, and capture photos or videos on command capabilities that are foundational in modern robotics, from autonomous delivery bots to inspection drones.
- RoboRide now offers a powerful combination of mobility, sensing, and visual awareness. It remains compact, affordable, and customizable.
- **RoboRide is where smart movement meets real intelligence.**



Project Overview

- **RoboRide is a compact, sensor-assisted robotic vehicle** designed for controlled transport and hands-on engineering learning.
- **Upgraded with ultrasonic distance sensing and a camera**, it prevents reverse collisions and captures photos/videos on command.
- **Stable and responsive motor control system** powered by PWM and Raspberry Pi for smooth movement in all directions.
- **Dual-purpose platform**, useful for educational robotics and small controlled-environment transport demonstrations.
- **Low-cost, modular, and easily expandable**, making it ideal for future upgrades like wireless control or autonomous navigation.





Project Purpose

- **Efficiency in Industry –** RoboRide is engineered to deliver a safe, reliable, and cost-effective solution for short-distance transport of goods within hospitals, warehouses, and campus environments. Equipped with a camera and ultrasonic sensor, it enhances navigation, obstacle detection, and situational awareness. By automating repetitive delivery tasks, RoboRide reduces human workload, minimizes errors, and significantly boosts overall operational efficiency

- **Creativity in Education** – RoboRide also functions as an educational platform that inspires innovation and curiosity among students. With integrated features such as its camera and ultrasonic sensor, learners gain hands-on experience in robotics, programming, and problem-solving. This practical exposure encourages exploration, creativity, and a deeper understanding of real-world technology applications.

Project Goals

- **Prototype Creation:** Develop a semi autonomous prototype with intelligent navigation support and a modular design for future upgrades.
- **Pilot Applications:** Showcase practical use cases in logistics, healthcare, and academic environments to test and refine capabilities
- **Ensure affordability and scalability** so RoboRide can be produced at low cost and adapted for different industries.
- **Engage students and educators** by introducing RoboRide as a learning tool in robotics, coding, and engineering.
- **Lay the foundation for commercialization** with a clear path toward market entry and investor opportunities.

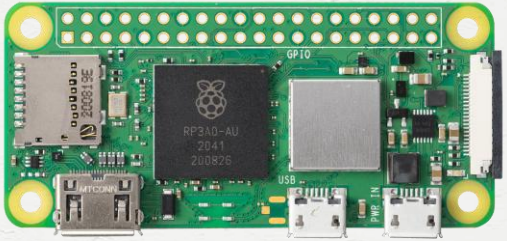
Technical Components

Ultrasonic Sensor (HC-SR04): Distance measurement and obstacle detection

Raspberry Pi Camera: For real-time visual capture

Raspberry Pi Zero W: Main processor and GPIO controller

Motor & Chassis: Provides movement and support structure



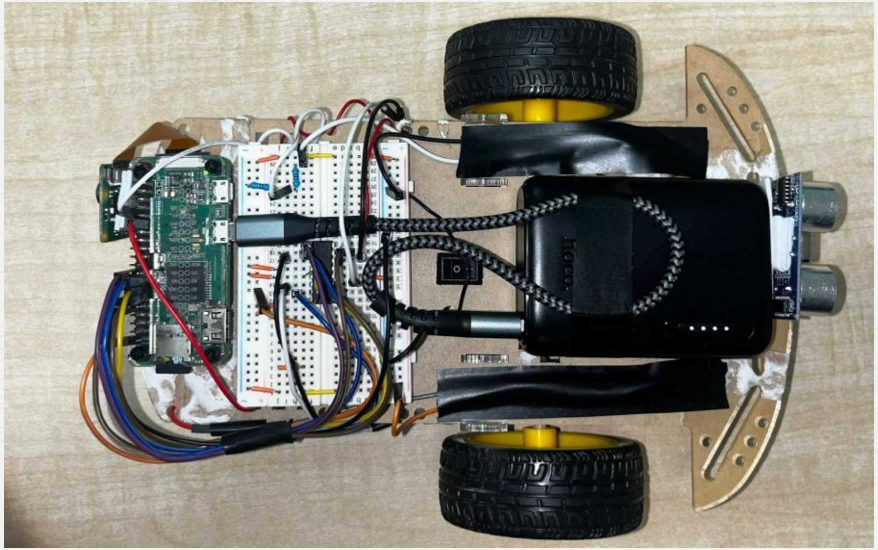
Bill of materials

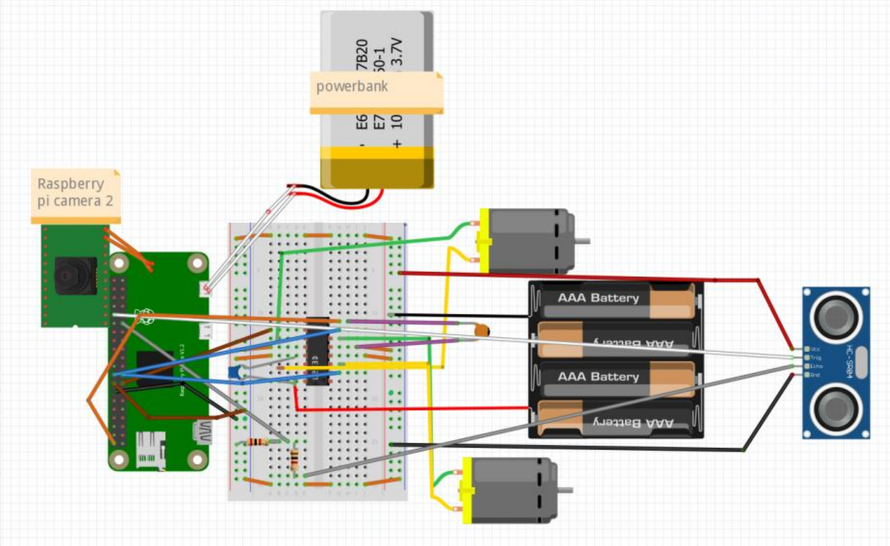
No.	Component	Estimated Cost (CAD)
1	Raspberry Pi Zero W	$60.00
2	L293D Motor Driver IC	$2.00
3	DC Motors (x2)	$10.00
4	2WD Car Chassis	$12.00
5	Breadboard	$6.00
6	Jumper Wires Set	$5.00
7	Power Bank (5000 mAh)	$20.00

No.	Component	Estimated Cost (CAD)
8	AA Battery Holder	$3.00
9	AA Batteries (x4)	$4.00
10	USB Cable (Micro-USB)	$4.00
11	HC-SR04 (Ultrasonic sensor)	$2.00
12	Raspberry pi Camera Module	$20
13	PI 0 W camera cable	$10
14	Miscellaneous Hardware (nuts, bolts, spacers)	$5.00

Total Price = $160 only

Assembly




powerbank
Raspberry pi camera 2
AAA Battery
AAA Battery
AAA Battery
AAA Battery
HC-SR04

Market Oppurtunity

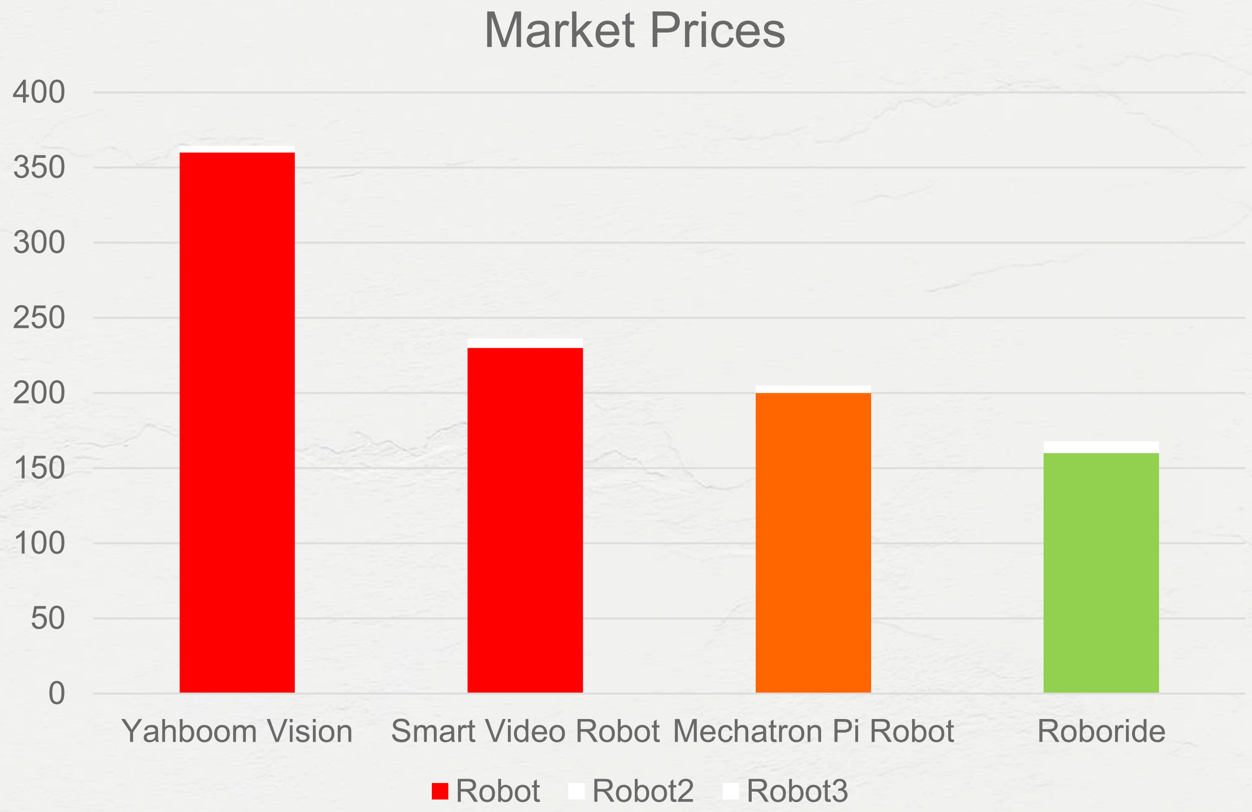
Applications:

- **Healthcare:** Ensures safe and reliable delivery of medication, lab samples, and supplies within hospital environments. Ensures safe and reliable delivery of medication, lab samples, and supplies within hospital environments.
- **Logistics & Manufacturing:** Transporting components across factory floors.
- **Education:** Used in schools/universities as a robotics teaching tool.

Unique Selling Points (USPs):

- Dual-purpose (industrial + educational).
- Modular, affordable, and customizable.
- High adaptability to multiple industries
- Easy Assembly.

Why RoboRide?



Investment Proposition

- **Dual Market Reach:** Serves both industrial automation and education sectors.
- **Low-Cost & Scalable:** Affordable to produce, easy to adapt for various uses.
- **High ROI Potential:** Low production cost (~$160) with strong profit margins.
- **Growth Opportunity:** Rising demand for smart delivery robots and STEM tools.
- **Call to Action: Invest in RoboRide — powering the future of smart mobility and learning.**

Challenges Faced and Solutions Implemented

- Faced difficulty connecting the Raspberry Pi and setting up GPIO pins correctly.
- Encountered motor polarity issues, causing the wheels to spin in opposite or incorrect directions.
- Wiring was congested on the breadboard, making troubleshooting and circuit organization difficult.
- Required multiple testing and rewiring sessions to stabilize connections and ensure consistent performance.
- Improved layout and cable management for a cleaner, more reliable build.

Conclusions

- RoboRide continues to embody the fusion of innovation, practicality, and education, but now with advanced safety features. Building on its strong foundation, Phase 2 introduces a camera system and ultrasonic sensors that enable the vehicle to detect obstacles and automatically stop at a safe distance while reversing.
- This upgrade not only enhances operational safety and efficiency in logistics but also expands its educational value by showcasing real-world applications of sensor technology, computer vision, and autonomous systems. RoboRide remains a scalable, cost-effective solution for smarter mobility while serving as a hands-on STEM platform that inspires creativity and innovation.
- **Invest in RoboRide — now smarter, safer, and empowering the innovators of tomorrow.**

Future Improvements

Add AI-based navigation

Implement IoT monitoring and mobile control interface

Improve battery efficiency and wireless charging

Support heavier payloads with modular chassis design

Thank you

“RoboRide isn’t just a robot — it’s a platform for innovation, learning, and smarter mobility.
Your investment powers the next step toward an intelligent and inspired future.”